

Euclid's Elements — Book XI

Proposition 23

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

To construct a solid angle out of three plane angles, any two of which are together greater than the remaining one, while the three are together less than four right angles.

1 Purpose and Scope

Proposition XI.23 closes the introductory trihedral block of Book XI. The preceding propositions provide the two necessary conditions for three plane angles to occur as the faces of a proper trihedral angle:

1. any two of the three angles are together greater than the third (XI.20);
2. the three angles are together less than four right angles (XI.21).

XI.23 proves the converse construction. Starting from three proper plane angles satisfying exactly these two conditions, one constructs a proper solid angle whose three face angles are respectively congruent to the given ones.

The production Lean theorem is

```
theorem euclid_proposition_11_23
  [Hinc : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [HSO : HilbertSpaceOrder
    (Geo := Geo) (H := Hinc) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := Hinc) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (A O B C P D E Q F : Geo.Point)
  (h12_3 :
    HilbertTwoAnglesGreaterThanAngle
```

```

      Geo A O B C P D E Q F)
(h23_1 :
  HilbertTwoAnglesGreaterThanAngle
    Geo C P D E Q F A O B)
(h31_2 :
  HilbertTwoAnglesGreaterThanAngle
    Geo E Q F A O B C P D)
(hFour :
  HilbertThreeAnglesLessThanFourRightAngles
    Geo A O B C P D E Q F) :
HilbertThreeAnglesHaveTrihedralRealization
  Geo A O B C P D E Q F

```

The final target is deliberately neutral. It does not expose the auxiliary circumcenter, the chosen base plane, or the normal construction. It asserts only the existence of a proper trihedral configuration whose three face angles realize the three prescribed angles.

2 Euclid's Statement

In the classical statement one is given three plane angles

$$\angle ABC, \quad \angle DEF, \quad \angle GHK,$$

such that every two together are greater than the remaining one and such that their total is less than four right angles. The task is to construct a solid angle bounded by three plane angles equal to these.

Thus XI.23 is the sufficiency theorem corresponding to XI.20 and XI.21. The logical pattern is

$$\begin{array}{c}
 \text{pairwise angle inequalities} \\
 + \text{ three-angle bound } < 4R \\
 \Downarrow \\
 \text{existence of a trihedral realization.}
 \end{array}$$

No numerical angle measure is introduced in the formal theorem. The two source conditions are represented synthetically by `HilbertTwoAnglesGreaterThanAngle` and `HilbertThreeAnglesLessThanFourRightAngles`.

3 Euclid's Classical Construction

Euclid first cuts six equal radii on the sides of the three given angles and joins the corresponding chord endpoints. By XI.22 the three resulting chord lengths form a triangle. Let this triangle be LMN .

A circle is described through L, M, N , with center O . The decisive metric claim is that the common source radius, say AB , is greater than the circumradius LO .

Euclid excludes equality by SSS: if $AB = LO$, then the three given angles would equal the three central angles

$$\angle LOM, \quad \angle MON, \quad \angle NOL,$$

whose sum is four right angles, contradicting XI.21.

He then excludes $AB < LO$ by laying off equal shorter radii on OL and OM , producing a smaller similar triangle. The printed proof uses VI.2, I.29, VI.4, V.16 and I.25 to compare its chord with LM , and hence to obtain strict inequalities between all three source angles and the three central angles. Their sum would then be greater than four right angles, again contradicting the hypothesis.

Having proved $AB > LO$, Euclid erects OR perpendicular to the plane of the circumcircle by XI.12 and chooses its length so that

$$OR^2 = AB^2 - LO^2.$$

The auxiliary lemma constructs this length by a semicircle argument using IV.1, III.31 and I.47.

Finally one joins R to L, M, N . Right-triangle arguments give

$$RL = RM = RN = AB,$$

and three applications of SSS recover

$$\angle LRM \cong \angle ABC, \quad \angle MRN \cong \angle DEF, \quad \angle NRL \cong \angle GHK.$$

Thus the solid angle at R has the required three faces.

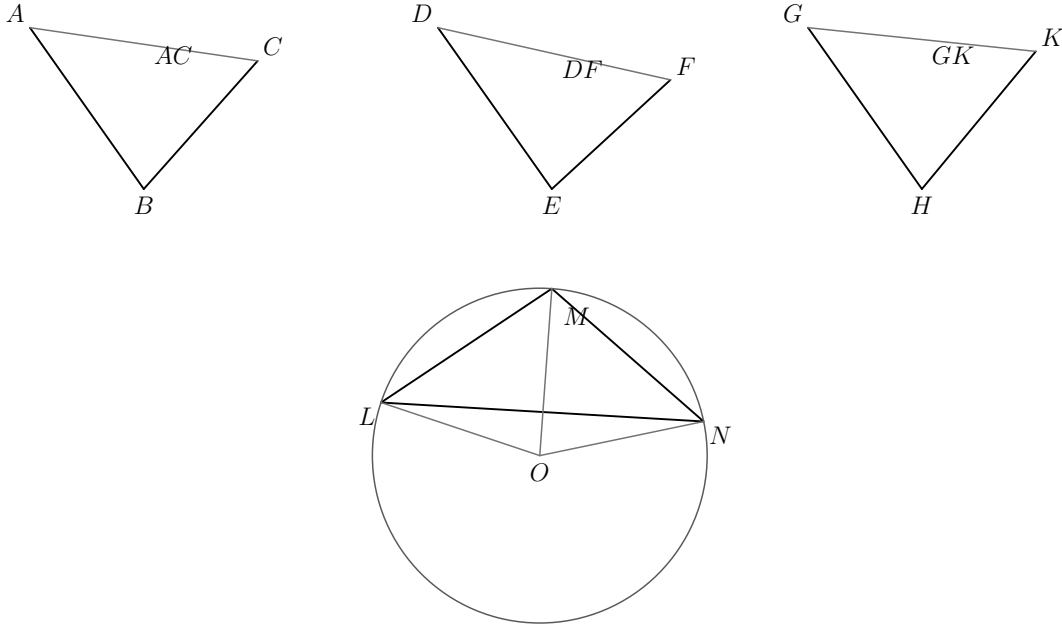
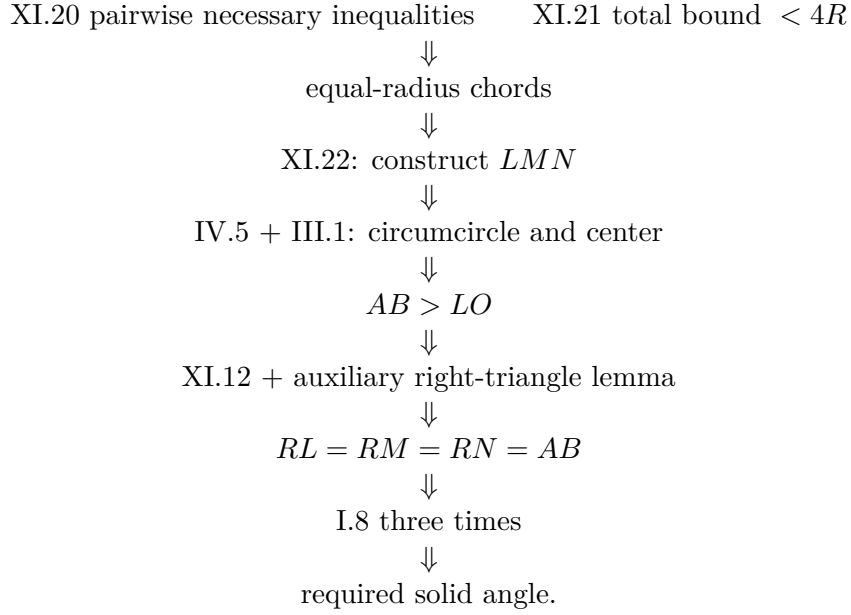


Figure 1: The first classical stage of XI.23. Equal radii are cut on the three prescribed angles; their three chords become the side lengths of the auxiliary triangle LMN by XI.22. The right panel shows its circumcircle and center O .

4 Classical Dependency Pattern

At proposition level the source DAG is



The long middle step $AB > LO$ is where Euclid uses the largest collection of earlier plane propositions, including the Book V–VI proportional route in the strict case.

5 The Formal Target in HilbertInterfaceXI

The Book XI interface contains the proposition-independent language

```

def HilbertTrihedralRealizesThreeAngles
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (R L M N : Geo.Point)
  (A O B C P D E Q F : Geo.Point) : Prop :=
HilbertTrihedralConfiguration Geo R L M N /\
Geo.AngleCongruent L R M A O B /\
Geo.AngleCongruent M R N C P D /\
Geo.AngleCongruent N R L E Q F

def HilbertThreeAnglesHaveTrihedralRealization
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (A O B C P D E Q F : Geo.Point) : Prop :=
exists R L M N : Geo.Point,
  HilbertTrihedralRealizesThreeAngles
    Geo R L M N A O B C P D E Q F

```

This is an important architectural boundary. The target records the mathematical output of XI.23 but none of its construction witnesses. In particular it does not require a global ambient `HilbertCongruence` instance merely to state the result.

6 Normalization into One Explicit Plane

The production proof does not assume that the three prescribed angles already live in a common plane. It chooses the plane π containing the second source angle and copies all three angles into that same plane with one common initial ray.

Schematically the copied angles are

$$\angle CPR_1, \quad \angle CPR_2, \quad \angle CPR_3,$$

with

$$PC \cong PR_1 \cong PR_2 \cong PR_3.$$

The helper

```
hilbert_XI23_copy_angle_equal_radius_in_plane_XI
```

performs one such copy. The general spatial transport layer then carries the three XI.20-type inequalities and the XI.21-type four-right-angle certificate to the copied angles:

```
hilbert_space_twoAnglesGreater_transport_all_in_plane
hilbert_space_threeAnglesLessThanFourRight_transport_all_in_plane
```

Only after this transport are the points repackaged as `PlanePoint` values and the four hypotheses converted to statements in `PlaneGeo( $\pi$ )`.

This separation is deliberate:

$$\text{ambient space} \longrightarrow \text{explicit plane } \pi \longrightarrow \text{PlaneGeo}(\pi).$$

No global planar geometry is installed on the ambient three-space.

7 XI.22 Inside `PlaneGeo( $\pi$ )`

Once the three angles have been normalized into the fixed plane, XI.22 is called entirely inside `PlaneGeo( $\pi$ )`. The result is a chord triangle, represented in the final normalized pipeline by

$$C, \quad R_2, \quad K.$$

The two nontrivial side data exported from XI.22 are congruences of the constructed sides with the appropriate equal-radius chords. The proof then establishes

$$\neg \text{Collinear}(C, R_2, K),$$

so the triangle has a circumcenter U in the same plane.

8 The Circumradius Inequality

The central planar theorem is

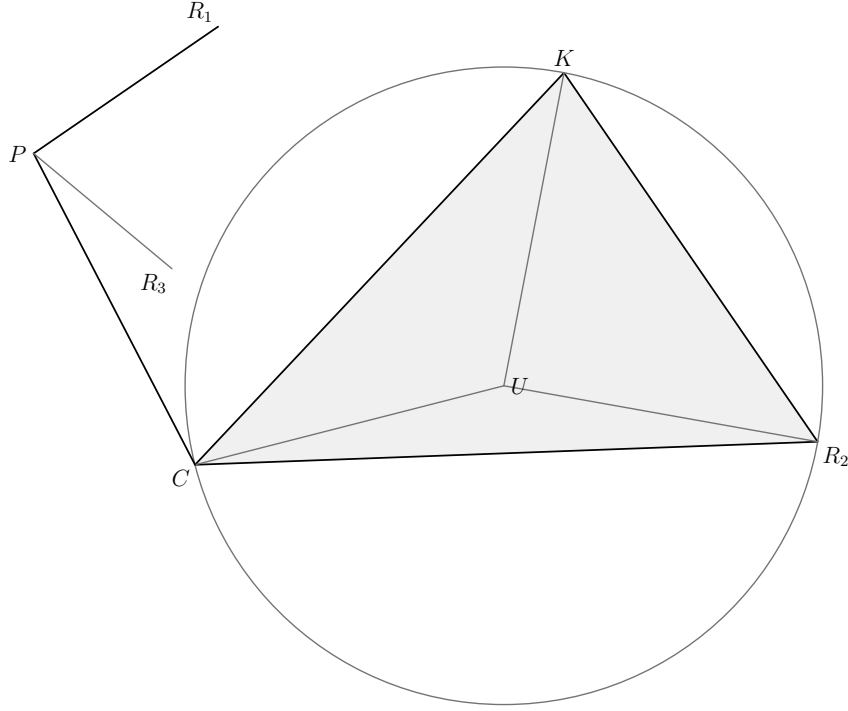


Figure 2: The normalized planar core. The three prescribed angles have been copied to one explicit plane. XI.22 produces the base triangle CR_2K ; its circumcenter U and equal radii remain entirely inside $\text{PlaneGeo}(\pi)$.

`hilbert_XI23_circumradius_less_source_radius_XI`

which proves, in the planar setting,

$$UC < OA.$$

The formal proof follows the same high-level trichotomy as Euclid but replaces the proportional Book V–VI argument by synthetic angle-order arguments.

First, equality $UC \cong OA$ is impossible. Equal radii would identify the three source angles with the three central angles by SSS, contradicting the strict four-right-angle certificate.

Second, the opposite strict inequality $OA < UC$ is impossible. The proof compares the corresponding equal-radius chords and central angles. Its main difficulty is not numerical computation but the possible cyclic order of the three rays from the circumcenter.

Finally, segment trichotomy leaves only

$$UC < OA.$$

A substantial part of the proposition-local proof exists to make this argument configuration-complete. The three-ray classifier separates the three “one ray enters the opposite chord” cases from the cyclic case, and each branch is discharged synthetically.

This is stronger as a proof audit than relying on a representative diagram: the conclusion is proved for every permitted incidence/order configuration covered by the theorem hypotheses.

9 The Auxiliary Height Segment

The theorem

```
hilbert_XI23_radius_height_in_plane_XI
```

packages the radius comparison together with the planar construction needed for the spatial apex.

From $UC < OA$ it constructs points E_0, K_0, H_0 such that the relevant right triangle has one leg congruent to the circumradius and hypotenuse congruent to the source radius. In particular the output includes

$$OE_0 \cong UC, \quad OH_0 \cong OA, \quad \angle OE_0H_0 \text{ right.}$$

The entire statement is made in $\text{PlaneGeo}(\pi)$. The ambient Euclidean-space structure supplies the Euclidean-plane instance only for this explicit slice. There is still no global **HilbertEuclideanPlaneGeo** assumption on the ambient space.

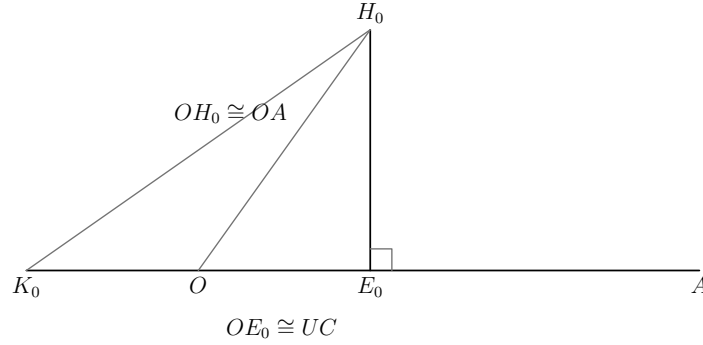


Figure 3: The synthetic replacement for Euclid's square-difference lemma. The planar construction encodes the source radius as the hypotenuse and the circumradius as one leg; the remaining leg is transported to the spatial normal.

10 The Spatial Apex

The normalized spatial package is

```
hilbert_XI23_normalized_spatial_apex_XI
```

Starting from the base plane π and its circumcenter U , XI.12 erects a line through U perpendicular to π . The height obtained from the auxiliary right triangle is laid off on this normal, producing an apex R with

$$R \notin \pi.$$

Spatial right-angle congruence and SAS then yield

$$RC \cong RR_2 \cong RK \cong PC.$$

The key point is that the final lateral edges are obtained without introducing coordinates, vector lengths, or a numerical Pythagorean equation. The role played classically by

$$OR^2 = AB^2 - LO^2$$

is represented by an explicitly constructed right triangle plus spatial SAS.

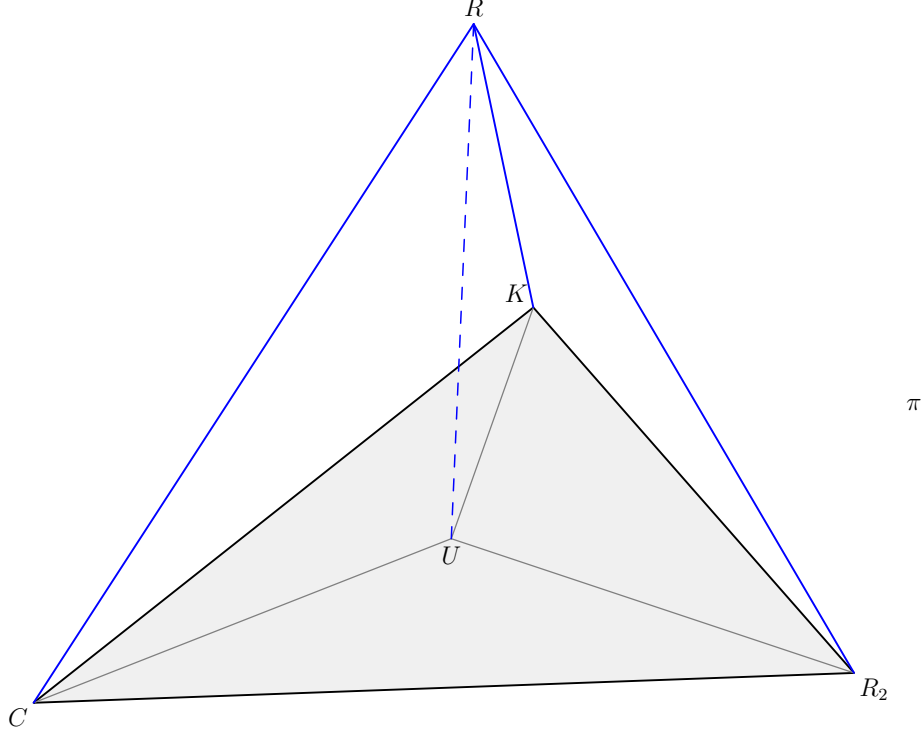


Figure 4: The final spatial construction. The base triangle CR_2K lies in π , U is its circumcenter, and R lies on the normal through U . The three blue lateral edges are proved congruent to the common source radius.

11 Recovery of the Three Face Angles

The final theorem now has three spatial triangles:

$$RKC, \quad RCR_2, \quad RR_2K.$$

Each is compared by SSS with one of the normalized planar equal-radius triangles:

$$PCR_1, \quad PCR_2, \quad PCR_3.$$

The common spatial theorem

```
hilbert_space_sss_angleA_in_plane
```

returns the required angle congruences across the ambient/plane boundary. They are then composed with the original angle-copy congruences to obtain

$$\begin{aligned} \angle KRC &\cong \angle AOB, \\ \angle CRR_2 &\cong \angle CPD, \\ \angle R_2RK &\cong \angle EQF. \end{aligned}$$

The cyclic order

$$R; K, C, R_2$$

is chosen because it aligns exactly with the three copied source angles.

12 Noncoplanarity and the Final Trihedral Package

It remains to prove that the three face triangles form a proper solid angle, not merely three planar triangles sharing a vertex.

Since C, R_2, K are noncollinear, they determine the base plane π . If another plane contained

$$R, K, C, R_2,$$

plane uniqueness would force it to equal π . But the constructed apex satisfies $R \notin \pi$. Hence the four points are not coplanar.

This supplies the `HilbertTrihedralConfiguration` part of the target. The final packaging is then purely structural:

```
hilbertTrihedralRealizesThreeAngles_intro
hilbertThreeAnglesHaveTrihedralRealization_intro
```

13 Hidden Formal Data

XI.23 exposes several pieces of geometric information that the classical diagram suppresses.

13.1 The source angles need not share a plane

The formal inputs are three ambient angles. Before XI.22 can be reused, the proof must construct congruent copies in one explicit plane and transport all four hypotheses to those copies.

13.2 The circumcenter must be certified to lie in the base plane

The spatial normal in XI.12 must be erected through a point already known to lie in the plane. The formal proof therefore reconstructs the circumcenter inside $\text{PlaneGeo}(\pi)$, rather than using an ambient circumcenter without an incidence certificate.

13.3 The circumradius comparison has genuine order cases

A single picture cannot encode all cyclic orders of the three radii. The Lean proof contains a three-ray classifier and treats the cyclic branch explicitly. This removes the diagram-dependent gap noted in modern commentary on the printed proof.

13.4 The final noncoplanarity is not automatic

Three correct face angles do not by themselves guarantee a proper solid angle. The proof uses $R \notin \pi$, noncollinearity of the base triangle, and plane uniqueness to establish the required trihedral configuration.

14 Relationship with XI.20, XI.21, and XI.22

XI.23 is best read as the converse synthesis of the previous three results.

- XI.20 : each pair of face angles is greater than the third,
- XI.21 : the three face angles are less than $4R$,
- XI.22 : the three equal-radius chords form a triangle,
- XI.23 : these data construct the trihedral angle.

At API level the same structure appears as

```
HilbertTwoAnglesGreaterThanAngle
      +
HilbertThreeAnglesLessThanFourRightAngles
      ↓↓
euclid_proposition_11_22
      ↓↓
euclid_proposition_11_23
```

The production proof therefore reuses the synthetic language developed for XI.20–XI.22 instead of introducing a new numerical theory of angle sums.

15 Book XI Interface and the New 3D Transport Layer

The final refactor separates three levels.

```
Hilbert3DAngleComparisonTransport
      ↓↓
HilbertInterfaceXI
      ↓↓
Proposition11_23
```

The first file contains proposition-independent spatial transport machinery: betweenness transport, ray/segment transport, subangle transport, transport of angle order, transport of the XI.20 comparison relation, transport of the four-right-angle certificate, and ambient-to-PlaneGeo bridges.

`HilbertInterfaceXI` remains the public Book XI facade. It owns the neutral trihedral-realization language and imports the generic transport layer.

`Proposition11_23` contains only the proposition-specific construction: the radius contradiction, the auxiliary height, the spatial apex, and the final assembly.

16 Comparison with Euclid’s Metric Route

The classical and formal proofs have the same large-scale geometry:

chord triangle $\longrightarrow$ circumcenter $\longrightarrow$ radius comparison
 $\longrightarrow$ normal apex $\longrightarrow$ SSS face recovery.

The main difference is the middle metric technology.

Euclid’s strict-radius contradiction uses similarity and proportion:

VI.2 + VI.4 + V.16 + I.25.

The production Lean proof instead uses the already developed synthetic angle-order API, equal-radius chord comparison, ray-order classification and the four-right-angle certificate. It therefore does not require a general Book V/VI ratio calculus merely to prove XI.23.

The second difference is the construction of the height. Euclid represents it by a difference of squares and verifies it by I.47. The formal proof constructs an auxiliary right triangle in the plane and transports its leg to the spatial normal. The result is geometrically equivalent for the purpose of the construction, but no numerical square operation is introduced into the Lean statement.

16.1 The Hilbert–Wyler route for XI.23

XI.23 now also has a production Hilbert–Wyler route in

```
CGJteamLab/Wyler/Proposition11_23.lean
```

with two final entry points:

```
euclid_proposition_11_23_wyler_core  
euclid_proposition_11_23_wyler
```

The distinction between them is structural. The core theorem assumes an explicit

```
[W : HilbertWylerAxioms Geo]
```

and routes the spatial incidence part of the construction through the Wyler interface. In particular, the proof uses the Wyler fields

```
W.plane_through  
W.line_in_plane  
W.plane_unique
```

for the choice of the normalization plane, preservation of points on lines in that plane, and the final noncoplanarity argument.

The public theorem does not add `HilbertWylerAxioms` as a new three-dimensional axiom. Instead it derives the package from the established Hilbert 3D environment by

```
hilbertWylerAxioms_of_hilbert3D
```

proved in

```
CGJteamLab/HilbertWyler3DCompatibility.lean
```

Thus, in the present XI.23 theorem, the dependency direction is

$$\text{Hilbert 3D} \implies \text{Hilbert–Wyler incidence} \implies \text{XI.23}.$$

Wyler is therefore derived incidence theory in the three-dimensional Hilbert environment, not a competing additional foundation.

Only the incidence routing is changed. The metric and order core remains the same synthetic machinery used by the direct proof: **PlaneGeo**, XI.22, the circumcenter construction, the radius comparison, the auxiliary right triangle, XI.12, spatial SAS, and the final SSS recovery of the three face angles. This is intentional. The Wyler theorem tests the incidence organization of XI.23 without duplicating an already established metric library.

The resulting theorem is genuinely different from merely aliasing the direct Hilbert theorem: the normalization and noncoplanarity steps are reconstructed through the Wyler incidence API. At the same time it is not yet a theorem of arbitrary-dimensional Euclidean geometry. No E^4 structure occurs in the production proof, and no local metric 3-flat inside E^n is assumed.

17 E^n Transfer Audit

XI.23 is not as dimension-free as XI.20 and XI.21.

The planar core—angle transport, XI.22, circumcenter geometry, chord comparison, and the radius contradiction—lives in one explicit **PlaneGeo** slice and is therefore structurally portable.

The new Hilbert–Wyler production theorem sharpens the dimensional audit. It shows that the incidence operations used by the 3D proof can be routed through **HilbertWylerAxioms**: construction of the base plane, line-in-plane closure, and final plane uniqueness. In genuine Hilbert 3D this package is itself derived by `hilbertWylerAxioms_of_hilbert3D`.

The final construction nevertheless still uses genuinely spatial metric data:

- a line through the circumcenter perpendicular to the base plane;
- a point on this normal outside the plane;
- spatial congruence and SAS across different plane slices;
- SSS recovery of the three trihedral face angles.

Consequently the project now has a Wyler-routed XI.23 theorem in Hilbert 3D, but not yet an arbitrary-dimensional E^n version. A higher-dimensional transfer would still need an explicit local three-dimensional carrier equipped with the required perpendicularity and congruence theory, or another theorem showing that the metric apex construction can be confined to a suitable 3-flat.

18 Axiomatic Status

The kernel audit was run on both the direct Hilbert theorem and the Hilbert–Wyler production route:

```
import CGJteamLab.Proposition11_23
import CGJteamLab.Wyler.Proposition11_23

#print axioms Geometry.euclid_proposition_11_23
#print axioms Geometry.euclid_proposition_11_23_wyler
#print axioms Geometry.euclid_proposition_11_23_wyler_core
```

Lean reports the same dependency set in all three cases:

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

This equality of kernel dependency sets is important. The Wyler proof is not an alias of the direct theorem: its spatial incidence front end is reconstructed through `HilbertWylerAxioms`. Nevertheless, neither the explicit `wyler_core` theorem nor the public theorem obtained from `hilbertWylerAxioms_of_hilbert3D` introduces any additional axiom. The architectural refactor changes the route of the proof, not its axiomatic cost.

The logical axioms `propext`, `Classical.choice`, and `Quot.sound` are part of the standard project baseline.

Two Book Zero convenience axioms occur:

```
Geometry.bookZero_nullSegment1
Geometry.bookZero_nullSegment2
```

The first expresses that a segment congruent to a null segment is itself null. The second expresses congruence of null segments. Both are already tracked project-level interface debt; they are not proposition-specific axioms introduced by XI.23.

The genuinely stronger geometric dependency is

```
Geometry.hilbert_circle_circle_intersection
```

This is the project’s circle–circle continuity axiom: under the usual inside/outside hypotheses, two circles meet. XI.23 therefore does depend, through its planar circle/circumcircle construction machinery, on a continuity principle.

This is an important distinction from XI.20 and XI.21. Those propositions could be expressed in the neutral angle-comparison and spatial-incidence layers, while XI.23 must actually construct the planar circle geometry used to build the base configuration and its circumcenter.

Accordingly, the precise status is:

New proposition-specific axiom:	no
New spatial axiom:	no
Book Zero null-segment axioms used:	yes, two
Circle-circle continuity used:	yes
Numerical angle measure used:	no
Coordinate or vector structure used:	no
Global ambient <code>HilbertEuclideanPlane Geo</code> :	no
Explicit <code>PlaneGeo</code> Euclidean slice:	yes
<code>sorry / admit</code> :	no
Full <code>lake build</code> :	clean

The continuity dependency is inherited from established circle-construction infrastructure. XI.23 does not introduce a new local continuity axiom of its own.

19 Diagrammatic Audit

Four geometric states are retained.

1. The equal-radius chord construction together with the XI.22 base triangle and circumcircle.
2. The normalized planar Lean core C, R_2, K, U .
3. The auxiliary right triangle representing the height construction.
4. The final spatial apex over the base plane.

No separate figure is introduced for:

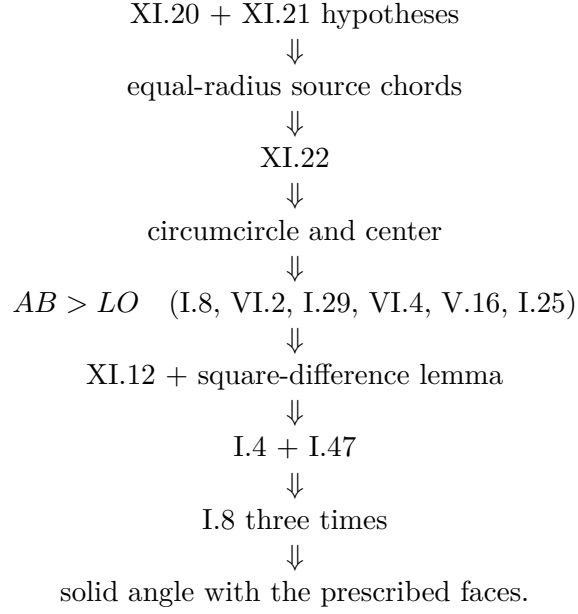
- each of the three angle-copy operations;
- each transport of an XI.20 inequality;
- each SSS call;
- the final plane-uniqueness contradiction.

Those operations alter logical information but do not create a new stable geometric state.

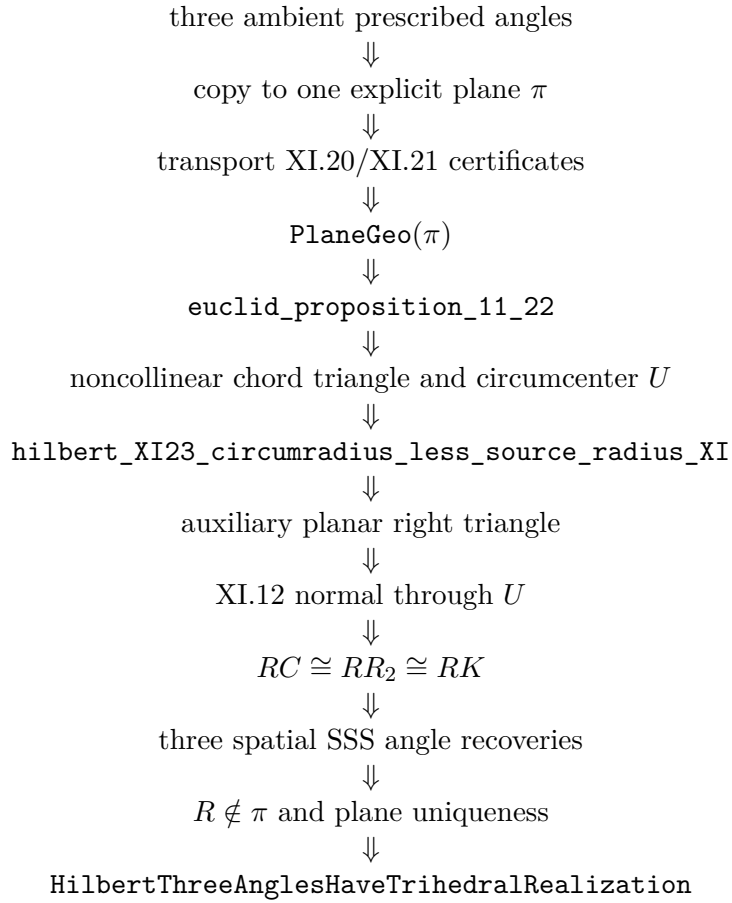
The figure convention follows the established Book XI style: planar geometry is black or gray; genuinely spatial edges are blue; a hidden spatial continuation may be blue dashed; plane fills are light.

20 Dependency Summary

The classical local chain is



The production Lean chain is



21 Conclusion

Proposition XI.23 is the constructive converse of XI.20 and XI.21 and the point at which the opening theory of solid angles becomes complete.

The final formal proof preserves Euclid’s global architecture remarkably well: construct the chord triangle, find its circumcenter, prove the common edge length exceeds the circumradius, erect a normal through the center, and recover the three required face angles by SSS.

At the same time, the formalization changes two important local mechanisms. The circumradius comparison is proved synthetically through angle order rather than through a Book V–VI proportional calculation, and the height is encoded by an auxiliary right triangle rather than by a numerical difference of squares.

The most important formal gain is control of hidden configuration data. The three source angles are explicitly normalized into one plane; the circumcenter is certified to lie in that plane; all cyclic ray-order cases are closed; the apex is proved to lie outside the base plane; and noncoplanarity is derived from plane uniqueness.

Thus XI.23 is not merely a long terminal construction. It is a useful integration test for the mature Book XI architecture:

$$\begin{aligned} &\text{synthetic angle comparison} + \text{PlaneGeo} + \text{spatial congruence} + \text{XI.12} \\ &\quad \longrightarrow \text{proper trihedral realization.} \end{aligned}$$